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#include <stdio.h>
#include "platform.h"
#include "xil_printf.h"
#include "xgpio.h"
#include "xparameters.h"
#include "xscugic.h"
#include "xil_exception.h"

XGpio slide, btn;
XScuGic_Config *gic_config;
XScuGic gic;

void slide_handler(){

    XGpio_InterruptDisable(&slide, XGPIO_IR_CH1_MASK);
    u32 dataRead = XGpio_DiscreteRead(&slide, 1);
    xil_printf("SLIDE SW Data Read : %0d\n", dataRead);
    XGpio_InterruptClear(&slide, XGPIO_IR_CH1_MASK);
    XGpio_InterruptEnable(&slide, XGPIO_IR_CH1_MASK);

}

void btn_handler(){

    XGpio_InterruptDisable(&btn, XGPIO_IR_CH1_MASK);
    u32 dataRead = XGpio_DiscreteRead(&btn, 1);
    xil_printf("PUSH BUTTON Data Read : %0d\n", dataRead);
    XGpio_InterruptClear(&btn, XGPIO_IR_CH1_MASK);
    XGpio_InterruptEnable(&btn, XGPIO_IR_CH1_MASK);

}

void gpio_init(){
    int statusss = XGpio_Initialize(&slide,XPAR_SLIDE_DEVICE_ID );
    int statusb = XGpio_Initialize(&btn,XPAR_BTN_DEVICE_ID);

    if((statusss & statusb) == XST_SUCCESS)
        xil_printf("GPIO INTI SUCCESSFUL\n");
    else
        xil_printf("GPIO INIT FAILED\n");

    XGpio_SetDataDirection(&slide, 1, 0xf);
    XGpio_SetDataDirection(&btn, 1, 0xf);
}

void gic_init(){

    gic_config = XScuGic_LookupConfig(XPAR_PS7_SCUGIC_0_DEVICE_ID);
    s32 status = XScuGic_CfgInitialize(&gic, gic_config, gic_config->CpuBaseAddress);

    if(status== XST_SUCCESS)
        xil_printf("GIC INTI SUCCESSFUL\n");
    else
        xil_printf("GIC INIT FAILED\n");

    Xil_ExceptionInit();
    Xil_ExceptionRegisterHandler(XIL_EXCEPTION_ID_IRQ_INT, (Xil_ExceptionHandler) XScuGic_InterruptHandler, &gic);
    Xil_ExceptionEnable();

    XScuGic_Connect(&gic, XPAR_FABRIC_SLIDE_IP2INTC_IRPT_INTR,(Xil_InterruptHandler)slide_handler, &slide);

    XScuGic_Connect(&gic, XPAR_FABRIC_BTN_IP2INTC_IRPT_INTR,(Xil_InterruptHandler)btn_handler, &btn);

    XScuGic_Enable(&gic, XPAR_FABRIC_SLIDE_IP2INTC_IRPT_INTR);
    XScuGic_Enable(&gic, XPAR_FABRIC_BTN_IP2INTC_IRPT_INTR);

    XGpio_InterruptGlobalEnable(&slide);
    XGpio_InterruptGlobalEnable(&btn);

    XGpio_InterruptEnable(&slide, XGPIO_IR_CH1_MASK);
    XGpio_InterruptEnable(&btn, XGPIO_IR_CH1_MASK);

}

int main()
{
    init_platform();
    gpio_init();
    gic_init();

    while(1){

    }

    cleanup_platform();
    return 0;
}
```